ActInf GuestStream 084.1 ~ "The Nature of Habits and Agential Systems", Jesse G

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all right hello and welcome it's May 30th 2024 and we're in act imp guest stream 84.1 this will be a presentation and a discussion with Jesse G so Jesse thank you for joining and for your participation looking forward to a great presentation and then we'll check the live chat and see where it goes from there go for it all right thank you so much for letting me do this showcase uh largely what I'm going to go over is basically an overview uh debut publication of a series of papers that have to do with uh habits and uh adaptive learning in regards to agential systems and basic basically uh looking at agential systems largely from uh the point of view of biological organisms largely trying to understand better understand connecting that with human behavior and largely the motivation here for this series of Publications it uh started with looking at a at a um sort of literature review that was Data focused on looking at the uh not necessarily addiction uh from a more broad point of view but from a drug dependency point of view when I reached out to a colleague of mine a number of years ago uh well little over three years ago now in 2021 when I first started uh this journey so so this uh whole uh this whole series of work has been undergone for about slightly over three years now so I started this work in 2021 originally I was connected indirectly with a group of graduate students uh at a state univ work uh associated with a State University and they were looking at data on uh on addiction um not not really broadly speaking but uh dealing with um looking at more so drug dependency based addiction and I was a bit I was quite disappointed with the effort put into the paper I mean I understand that the the students involved were uh were uh students in data science and informatics so but at at the same time I was a bit dismayed by the approach in the paper and I provided a critique of their work and I thought to myself well wow like addiction has some root there's some root to addiction we need to better understand and and what's also uh what was also um came off quite um odd to me for a number of years now is just the The Narrative looking at the lens of like what is the root of addictive behavior and that kind of goes back fun know thinking about fundamentals um and fundamentally we we think about habits and when people think about habits uh mostly stereotypically again uh people think about habits in sort of a a bad point of like you know habits are bad you know point of view and again reapo fundamentals and breaking the stereotypes [Music] um I think it's important to understand okay well our habits good or bad what can we say about them and how do we adapt to circumstances that we face as as humans you know as human beings um more specifically uh thinking about about um sort of uh this idea of not necessarily coping but rein reinforcing our fitness uh in the sense of like how can we better engage with the world around us and with each other and ourselves most importantly over time over the course of a lifetime uh and um so that's been the general focus of this series of of of works is trying to look into uh in terms of adaptive learning um what interventionary approaches like what primary interventionary interventionary approaches which I'll get into a little later um can be leveraged to be reinforcing towards our behavior and compliment how we engage with the world around us each other and ourselves rather than be counterintuitive um and looking at is sort of like kind of like an EB and flow uh of what where like how we drive how we how we motivate and direct our behaviors um so habits can be good or bad but more so thinking about habits on a spectrum of being reinforcement based in a sense that it complements our uh our fitness and how we engage with each other in a constructive way in the world around us of course uh as well as our our El uh rather than counterintuitive or maladaptive um so um anyway I didn't really give a former a more formal introduction of of who I am I wanted to kind of give start with a bit of an overview of motivation and and drive of the work I'm doing um so uh I call myself an independent researcher I'm working on a portfolio uh of of work where I spend quite a bit of time investigating academic Li literature um I I had for for this series of Publications I'm working on I have I have roughly about uh close to a thousand maybe little close to a thousand papers that uh to work with um I have hundreds of papers that I've I've I I found that uh I've read abstracts for that I haven't given a good read yet but I but I read quite I've gone through quite a bit of papers and um it gets to a point where you have so you find so much work to work with that you kind of have to figure you kind of have to prioritize okay do I have enough content um and um anyway um so basically what I'm trying to do with the portfolio work I'm doing is I I I look for data um I also find a lot of literature so I think it's important to find not just data but information criteria that tells a a story about the data that's out there um and I I I could go on a tangent about uh my you know the limitations of data science oriented oriented communities not really telling much of a narrative rather than looking at the data so what what I try to do with the work I'm doing is um I try to find a lot enough criteria which I consider to be information or or literature so I find a whole lot of criteria that emphasizes on on model building and construction not just looking at data but um data is important and um but more importantly like connecting uh you know building models not just about the data but being able to uh tell a story about the direction in which if we collect data in a sufficient manner where that would go where you know where you know uh the models would tell us uh like you know what does the information tell us in terms of that evaluates the criteria of data and the models that we that we build to have an understanding Narrative of uh um you know uh what we're working with in terms of uh uh in this case what I'm working on the series of papers dealing with habits and adaptive learning uh really getting to the root of the problem domain that we're working with here um and uh so basically what I'm trying to do again is um you know build models M mainly complex systems based models and um work with whatever data I can um to to to provide a narrative and and provide a means to justify okay this is the problem domain we're looking at this is how we should evaluate based on the the criteria or the the literature that has that has elaborated on on on the problem domain take the converging aspects of the literature so in this case looking at habits and and adaptive learning um and um you know providing emphasis of like okay this is where we can go based on the problem domain so that's B essentially what I'm trying to do whether or not um you know I end up working in Academia or industry in in some way or another um what I feel like I'm doing is more analytical like analytics based um not necessarily engineering focused although uh I am working on systems based approaches in terms of um interventionary uh um means to deal with adaptive learning and habits which is not the current paper I'm I'm elaborating on although um that's that's going to be the next paper but um I might get into that a little bit uh a little later but in a nutshell um even though that might be a bit of an elongated explanation about who you know who I am what I'm trying to do um I'm also trying to be agnostic about explaining what I'm trying to do not like even though I feel like I have an interest leaning into the cognitive Sciences given that's largely the uh the arena that this this paper is is is leaning is geared towards is is the cognitive Sciences domain and Ag gental Systems um I don't want to limit myself in in a corner per se I I I like to be open about my capacities of of building models looking at criteria looking at data and um provide in uh you know an evaluation of problem domains we are investigating anyway so the whole purpose of this talk now um is to go over my debut publication work which is an over more of an overview of understanding habits and adaptive learning with regard to agential systems uh there are four primary paradigms of interventionary approaches to Leverage adaptive learning uh with respect to human behavior U I'll go over that a little bit U maybe towards the end uh of the points I'm making but um what I plan on doing um I'll present a a base model that I put together uh and uh momentarily but I'm going to step through the paper a bit because there's there's a lot of context um but um I'm going to try and summarize the uh the points that I make I um for those of you who have looked at my work um I will say that uh it is quite a long paper so there is this metaanalysis approach to to the work in in terms of uh you know looking at a lot of literature that talk about habits and uh talks about Ag gental Systems and different models of of understanding um perception action uh perception based Dynamics and and we engage with the world around us and how we anticipate uh you know how we anticipate how to to leverage engagement over time based on our history of Engagement so anyway let let's let's start with the abstract and and we'll we'll we'll build up even further so uh like I mentioned uh this paper presents itself as an overview of looking at uh understanding the nature of habits and adaptive learning with respect to AG gental Systems so we can uh the best way to understand what a habit is is a persisting composition of behaviors that uh that are that are that couple our our engagement to to to leverage um how we reinforce uh uh interaction with our envir with our surrounding environment and um with other agents as as well as ourselves right um so on on the one hand uh agents can engage with their environment in a way that's constructive uh as well as each other and themselves uh so we call that reinforcement learning right where uh agents are are engaging in in a way that's constructive to optimize Fitness probably buffer in the best case buffer their Fitness um but uh on the other hand uh agents also can engage with their environment in the means that's could be deceptive or counterintuitive that that could not only deceive themselves but other agents in in in the environment right uh so that has to do with Mal adaptive learning so we can think of uh reinforcement learning being on one side of a spectrum and maladaptive learning being on the other side of a spectrum right and and we can say realistically that uh there's no truly completely uh you know uh there's no certain behav you know there's there's not no there's no most optimal Behavior that's going to completely optimize the fitness of an organism or or as as as a multi-agent systems um given uh the problems with cell cellular syence and aging that over time we engage with the world around us more suboptimally but um The Challenge there being that U to optimize Fitness in the face of uh thinking of of of aging and cellular inessence uh being a problem uh that we don't you know we're mortal beings agents don't last forever we break down um we decompose right so um um in in the face of of that issue of like becoming more suboptimal over time reinforcement learning uh leverages optimizing in in in the and as time goes on while U maladaptive learning is is really working against um the the the optimal means in which we can leverage Fitness um so the paper is pretty comprehensive and and the reason why it is is a bit long is one like I mentioned it's more it's looking at a whole bunch of literature connecting ideas uh and uh um and uh so I tried to approach the the content the the textualization of of of of the paper in a way that um is both comp you know comprehensive in terms of uh the audience that I expect to be a more General General audience to to be to be reading it um I'm I'm not too confident in terms of like how many actual cognitive scientists neuroscientists or physicists or whatever are going to be looking at the paper as opposed to more Layman type so I try to be mindful about who's looking at or who is who's going to be reading the paper and I try to be comprehensive as much as I can so there there is some redundancy here and there in terms of my elaborations on on specific points especially talking about perceiving acting and anticipating uh the world around us like this idea of what I call the pap cycle which I constantly bring up uh in in my work uh well in this particular paper so we could think of the pap cycle as perceiving acting a perception action priers so perception has to do with um so before elaborating on what I'd call the pap cycle um so as agents we're constantly like engaging with stimulus so any any anything that uh anything that provoke anything that provokes engagement for agents so agents we can think of as um single so we can think of the scaling of Ag gental Systems to be on the one hand units of Agents or single agents H so like single celled organisms you know your bacteria your and different types of cells and all that stuff like simple cells that don't have much degrees of of freedom in agency because uh they they they have simple functions and they serve simple roles in the in in their in the environment that they're embedded in which will get more into this this idea of how agents are existentially bound bounded to an embedded environment in in which they co-inhabit uh so now on the one one hand you we have these single agent systems that have simple functions and uh they serve a certain role within the environment they're embedded in and on the other hand we have compositions of single agents that make up what uh uh make up a multicellular system so in terms of speaking about agency I'm largely going to talk about and agential systems in general I'm largely going to focus on biolog iCal systems and and to be frank and I do kind of emphasize this maybe somewhat bluntly uh in my work uh uh realistically our most comprehensive and intuitive understanding of Agents really is reflective in biology in terms of uh um you know uh there they really uh you know you could say that there are very simple synthetic uh agential systems uh being uh uh researched by groups like uh Neil Gershen feld's Lab at MIT for instance uh Michael Levan uh Michael Levan's lab uh also investigates uh uh xenobots and and and what have you but but but Michael Lev's work is of course more is focused fed on biological organisms as agents um so um so I guess the one crutch of understanding Ag ental Systems really is that we're kind of limited to the scope of understanding Ag ental Systems in the scope of biology um so that's kind of the framework that that I'm focused on here even though I do pay a little mind of like comparing okay well computer systems that we' built using techn our understanding of Technology uh well very briefly I I do touch on that uh as opposed to what we consider to be Ag gental Systems um now um okay so what separates a computer you know the a computer system from an agal system um is that uh uh we are constantly engaging with stimulants as I as I mentioned uh now stimulus is is really uh you know anything that immediate that that that is directly you know anything that directly provokes engagement of the agent right so like whether it's something visual whether um that deals with any of of the Capa any capacity any modality for the agent to to to to have a capacity to develop some kind of sensation as as a result of being uh being provoked by the stimulus where like you could think of the stimulus as like B like constantly bombarding um uh the the agent in question right like the stimulus kind of is like this continuous input uh that um where the agent has multiple modalities of trying to process that input as Sensations and we're going to talk about what I call generalizations uh generalizations of Sensations so there's kind of multiple layers here in which Sensations are are processed by the agent uh so in in line with what I call the pap cycle uh perceptions actions and priors so perceptions deals with this uh capacity of of enveloping a p perceiving the stimulus so so so so there's this uh initial there's this layer of perceiving stimuli um where the agent again you know is you know is constantly um uh you so so so the agent within this embedded environment that it's existentially bound to is constantly being bombarded by stimulus right there there's constant uh uh constant uh exposure to to stimulus stimulus is inevitably uh uh uh directed towards the to towards the agent and then there's multiple modalities in which the agent then tries to process the stimulus as the stimulus is directed towards the agent and and the agent has a various number ways to try to process that stimulus and what and we can think about those modalities in the human sense as like our different senses so Vision touch uh auditory taste right um we can think of like like the combination of sensory modalities as a means to try to generalize uh uh how to process these Sensations in a visceral in so in so the generalization of Sensations aspect is this visceral initially this visceral processing of Sensations um so so you could think of like this visceral processing to be the the the initial layer of which that um the agent tries to process uh these Sensations trying to like this is kind of where the agent filters uh what what what it would try to try to um try to uh to distinguish between what is noise in the environment like what what the agent should should or should not prioritize attention to uh as opposed to what what could be significant to the agent like what what what what are what what sorts of stimulus that are being fed into the agents um modalities of processing um you know what stimulus matters and what stimulus doesn't matter um so this so so through the through through the generalization of Sensations there's this filtering of okay what what matters and what doesn't matter based on my circumstances of of leveraging Fitness as an agent Moment by moment um you know how do I handle that as an agent uh so and um going from there once the agent is able to prioritize once the agent recognizes certain patterns of stimulus that matter then there's this there's this means to categorize that stimulus as information so so so now then uh from there the agent might additionally try to Prior prioritize that information okay what information matters uh you know there might be redundant information or there might be information that matters to to my fitness right so we we can think about how that the information now that we have that is that went through that General that layer of being generalized as a sensation and differing differentiating trying to trying to understand understand whether that sensation is just like noise that can't really be pattern you know we can't be categorized as as something that might be redundantly meaningful in some some sense or it's something recognizable to the agent but not meaningful um or uh okay right so so from this generalization of Sensations we get to the point where um the agent is trying to figure out uh trying to figure out uh if there's a pattern to distinguish you know between okay well is this just just noise that I can't really comprehend and is therefore not meaningful or is there some pattern here that provides some information about my environment and my circumstances and then by providing some okay okay now I have information now I have this generalization of information um now I speak of the term generalization because that kind of leads to the problem of uh what I like to call Ver veridical perception that um and a lot of a lot of uh a lot of a lot of uh researchers and what have you talk about illusionism that in the sense that um we can have a more General a more we can develop a a gen we can always have a generalization that it e either is more becomes more intuitive or or becomes counterintuitive to how we leverage circumstance but we never have an absolute understanding of the information and and the uh as well as our experiences uh which is which is sort of a problem with with with philosop in the philosophy of science with okay well how do we evaluate something to be a theory as opposed to a conjecture which really has to do with with the rigor involved and what have you but um the problem with veridical perception is we are always constricted to a relative understanding of reality in which um an average an you know if if if enough agents agree with a a generalization then that generalization guides a you know a sufficiently large group of Agents so uh we can think about the direction of how we uh develop Technologies and how modern civilization got to where it is today and we're going to talk more about inter agent Dynamics uh momentarily but um the idea is that um we're constantly generalizing world around us we and uh there there's and and even though we can come up with generalizations which many which most of us could largely agree with those generalizations could be challenged by uh looking through a different lens of of generalizing that criteria so it's sort of a paradox of like okay well how do we evaluate the Cris criteria that we have as knowledge when you know we we have sort of these best guesss so to speak of of understanding uh what's happening with the world around us and how we understand reality and how we engage with each other um how we develop Technologies uh but um anyway um okay so going back to understanding habits and and what they have to do with um with Behavior Uh again I talk about uh you know I I talk about habits being these persistent compositions of of behavior and um on the one hand um we have Dynamics pertaining to Nature and d uh of of the agent and the nurturing of the agent right so we think about nature versus nurture now um so nature pertains to phys physiological limitations while uh nurture pertains to the exposure over time so we could think of like the you know the course of a lifetime an agent is exposed to environmental circumstances and um there's this idea of intrinsic and extrinsic variability of circumstances of of environment so uh we can think on the one hand that as multi-agent systems being human beings um we uh we we are a whole ecosystem right we are composed of multiple cells and we have these different anatomical layers uh that uh that allow us to have this integrated functionality in which we uh leverage metabolic and Cog cognitive low uh now the key thing to understand here about metabolic and cognitive load is um is top is the top down as opposed to bottom up Dynamics in which agents facilitate their own Integrity um as as a m um and and we're speaking largely about multi-agent systems because um uh there's this complexification of behavior that enables the capacity of for instance our ability to make decisions because we have a central nervous system not all organisms have a central nervous system right um so we especially when we think about the single agent systems they're much more simple in function they have much lesser degree of of agency since we're composed of multiple cells and all these different subsystems that regulate functioning uh that deals with metabolic cost um that gives us a bit more degrees of freedom because there's all there's all these these these different subsystems engaging with each other that facilitate the uh the capacities we have to engage with the world around us so on the one hand uh thinking from bottom up Dynamics uh we have the the composition of cells that compose of different organ systems right that uh regulate metabolic load um that enables the possibility from the top down for when we think of cognitive load from the top down we're thinking about executive functioning um so so from top down Dynamics uh we're thinking about um how cognitive load is facilitated where decision making happens um and this we we'll talk more about uh the the uh what is um fundamental to this decision making momentarily U but um thinking about nature Dynamics we're thinking about the physiological limitations of of the Integrity uh of bodily functions from the bottom up that influence the top down Dynamics uh that facil itate cognitive load leverage right so so so problems with physiology uh has has to do like maybe maybe with with uh problems with with with uh you know from from the beginning of of the agent's life cycle or at some point there's some hindrance that that sticks uh that kind of sticks out as as like a barrier for the agent to to leverage um how they engage with their environment whether or not you know let's say uh uh an amputated limb for instance like I you know like like I like if I lost my arm like how is that gonna impact how I engage with the world around me right or something more severe than that right so on the one hand we think about physiological limitations pertaining to nurture Dynamics uh we can think of like the the the beginning life of the life cycle an agent like considering birth problems problems from birth or we can think think of like something that just happened that could be an existential hindrance that could be permanent or or pretty long lasting until that that hindrance is corrected um whether or not it could be corrected and and in terms of nurture Dynamics since nurture Dynamics have uh uh uh you know pertain to the course of a lifetime exposure uh in which an agent leverages how uh it interacts with its surrounding environment other agents and intrinsically and extrinsically of course thinking about how the agent deals with its its own uh uh its own internalizations and externalizations as well as considering how it engages with other agents and it surrounding environment um and and the nurture Dynamics have uh generally would have the most impact because it deals with the full life cycle of the agent right so like you know until the agent ceases to exist it's going to be constantly engaging with an environment which it's bound to and it's constantly going to engage with other agents and and the the exposure Dynamics are are what are what is Central to this engagement and um both intr intrinsically and extrinsically right um and now um I was talking about metabolic load um I and uh now uh like every function of an a of of the agent um like every every process that happens like every interaction you know when we think about this you know from the Single Cell level to the multicellular level of of the um Integrity of of different functions of the agent from the bottom up from sorry from from from the uh um the top down to the bottom up um all that involves metabolic load and I even argue that the the process of of having thoughts that that like even generating thoughts in our head even in in a way that's compulsive or you know thoughts that that just kind of go in and out or or you know other processes like that involve perception action and priors all that comes down to metabolic load processing um and and those metabolic Dynamics are are facilitated through you know um are largely uh facilitated from the bottom from from you know from the bottom up um you know in terms of of of of of the integrity at at scale um okay so let's move on to routines um being fundamental so rout rou Ines uh we can think of uh R routines like you know for those of you who are programmers or or have a programming or engineering background we can think of a routine as like some kind of instruction or or or or or some kind of a a simple instruction among other uh instructions that enable Behavior to happen right and and we can also think of routines as uh compositions of uh uh you know this this continual uh we can also think of routines as the enabling of persistence to happen right so on the on the one hand you know we can think of routine as uh you know part some aspect of the Behavior Uh on the other hand we can also think of U more more more so I contextualize routines to be um the means in which the persistence uh uh of a be of of some compositions that make up a habit right to be to to be able to be orchestrated because the because a habit you can think of of a habit to be some routine uh um some routine characteristic of which the agent is act is leveraging um in some procedural orchestrated fashion right um that that some that that either um uh constructively optimizes its Fitness or or maintains its its its current state of Fitness in a way that right when we think about optimizing on the reinforcement aspect uh re terms of reinforcement learning or being counter you know routines and uh like this routinization of behavior of habits um you know could could be counterintuitive to the agent too right um like this Persistence of of behavior could be counterintuitive uh to to the behav behavior of Agents as well and and and and we this regulation uh of uh now now this regulation of of behavior uh we we we can think of is not not static right you know like it you know we're not consistently in the same state when we're engaging with our environment because the environment that we are embedded in is not in a fixed State either so the environment as well as ourselves as agents are not in a static state so we're like these D we're these complex uh adaptive dynamical systems complex in the sense that uh we have you know we're composed of uh uh you know like I mentioned like we're we're composed of multip we're these multicellular systems where we're a whole environment ecosystem have you but we're we we are composed of all these integ integral functions that regulate an entire system um so that's on the complex complex side um and then adaptive being that system is trying to compensate a a way to regulate Behavior based on circumstances right whether or not the circumstances uh optimize or or deceive the organ or deceive the agent so that that that's that's what what I mean in terms of thinking about uh comp you know agents as complex adaptive systems that uh you know know we're we are these these compositional systems that uh have certain um exhibit certain behaviors in a means that try to regulate uh our capacity to engage with with our environment and in the sense of you know thinking intrinsically uh regulating our own our our own bodily functioning and cognitive functioning as well as you know extrinsically as well how we engage with our surrounding environment um as well as other agents because those other agents are part of the surrounding environment and we're part of the environment as well hence we're we are embedded in in this environment right so um obviously we can think of this uh you know the reinforcement learning as opposed to Mal adaptive learning capacity of agents to be like sort of a tug OFW um you know hustle in terms of like optimizing as opposed to deceiving Fitness right because on the one hand there there are moments in which we can optimize leverage in which we can strategize certain mean certain you know ways to reinforce our behaviors and and those ways could themselves become counterintuitive over time based on how circumstances might change uh because the environment is not in a static State it's it's in this you know we're in this we exist within a complex system as complex systems right that are are are are are changing State over time due to to due to um uh you know the you know the state uh of which the uh entire environment you know like what you know the resources in the environment available to agents to to utilize you know whether or not how do resources remain available how those res you know uh those resources might change and and how other agents might change their behavior or conduct themselves as well as ourselves um and so you know we're we're always trying you know as agents we're always trying you know we're we're we you know I don't want to use the word exploit as a bad word but we we are uh trying to figure out a way to leverage resources and you know you know what can we use as resources in our environment optimize metabolism you know to to optimize our bodily functioning capacities from the bottom up to the top down as well as uh this idea of uh utilizing certain aspects of the environment as tools and and developing Technologies and communicating with with other agents um as as well well uh all right so as agents we we have this gold directedness and orientation and this this routine Behavior like this like the persistent dynamics of habits and just kind of like thinking of of of habits as like like these reoccurring behav ways um or uh these ways that we repetitively engage with uh with uh with our environment and other agents and even ourselves the way we kind of engage with our ourselves and like how we start our day and how we end our day how we you know we work on tasks and all that uh thinking about human behavior um uh so and and that is like you know there that's aligned with uh you know this engagement is is in line with uh you know this you know when we think of goal orientation we're thinking about uh you know how an agent kind of facilitates you know thinking about okay I'm going to do this to to achieve this or or or get this resource you like I want to have this resource I have to to do a certain number of things um so goal directedness could be you know composed of a subset of goals or there's change in in you know there are dynamics and you know the goals themselves are not necessarily static but uh in terms of like instinctual goals like we have to maintain metabolic load sufficiently for for our integrity right so so we can think of like there's certain certain goals that are are primary that facilitate um how we optimize Fitness like you know eating eat you know dieting uh exercise you know well thinking about human behavior we think about like stuff like dieting exercise sleep uh drinking water hydration like what what you know what are what are fundamental uh fundamental goals that are going to regulate inte you know this integral uh integrated um uh functioning uh from bottom facilitating bottom up and and top down facilitation in which an agent leverages uh optimizing Fitness right so so when we think about gold directing this we're really thinking we're really talking about okay how how is the agent uh optimizing its Integrity you know keeping itself alive really is is is like most fundamental to instinctual goal directiveness and there's there there can be other goals that complicate the agent's Integrity like obsessions and and like and how obsessions manifest and so you know like we can think about about oh right okay well as humans like some of us are into sports some of us are into video games or or or what have you or you know I like to to you know I'm I I I like to draw art art sketches so there's certain things that are you know there's certain other goals that might not necessarily um be directly or or or maybe their tertiary uh to to to the agent right there's certain goals that might not necessarily uh uh be be necessarily like I don't I don't necessarily have to work on artwork you know I could do I could F do other things but like goal alignment especially in terms of like goals that are not necessarily uh uh uh uh vital to to to the agents Integrity like when we think about like food water sleep and what have you right those are fundamental goals like there there's there's also the complication of other goals that uh kind of align with the interests of the agent over time based on exposure Dynamic when we think of nurture Dynamics like you know like what what stimuli ends up directing the agent to develop these interests and obsessions and what have you and now they they have these goals that are that are aligned like those obsessions or there or those interests or what have you not to say that interests are obsessions but um just trying to come up with some examples there okay um so I I kind of go uh I I I I uh in this section over here talking about um routine Behavior being integral to to environment and agent based Dynamics I kind of start off with like talking a little bit about uh you know from from the perspective of of of thinking about physics and particle interactions um and then getting to uh of agent based behavior um so we can think about how systems might um how how do how do how do how do systems how do agents as as systems uh maintain stability or acknowledging of course the problem with cellular syence and aging of systems like that system degrade over time um due to uh you know uh systems degrade over time after prolonged exposure toh to their environment and and what have you and and those Dynamics which see which you know I don't think it's necessarily necessary for me to talk about entropy and what have you but most systems you know uh degrade from a a a a structured uh more organized state to a more uh decomposed chaotic State um which is which which that that kind of connects in a sense to to to problems with cellular inessence like how as we get older our fitness is more suboptimal but even though our F Fitness is more suboptimal as we get older we still try to you know optimize uh optimize the Integrity of of of our agency by by engaging with with our environment in ways that are are are reinforcing reinforcement base even in light of that suboptimality as opposed to being counter inuitive uh to Fitness let's see um and now um now resources I mentioned resources uh re resources provide a VAR A variation in which we can leverage our fitness right so uh resources could be like food that allow us to manage manol load they could be uh certain aspects of the of the environment that we can use as like as tools to manipulate uh the landscape of the environment um so variational constraints in terms of optimizing Fitness uh in line uh of of our agency have to do with not how we uh maintain ourselves but how we engage with each other especially as a means to to maintain uh our fitness and there and and the environment can not all not only presents resources that agents can utilize but there might be barriers in the environment there might be optimal there in in a sense that there's obstacles within the environment that provide constraints in which the agent has has has certain agency to engage uh to engage with the environment um more more intuitively or or you know to to have better circumstances as opposed to circumstances being being worse uh based on certain resources not being available to to the agent so that could that could be an obstacle whe like whether it's there certain resources that that an agent was utilizing whether it has to do with metabolic uh metabolic uh Integrity more directly in terms of food or whether there are there's certain aspects of the environment that can be utilized as as as like as tool construction or means to communicate uh with other agents right um so so the environment um you know can pose as a barrier if it can make it if it if it if the environment um makes it that it's more difficult rather than more intuitive uh for agents to uh facilitate uh Integrity uh their Fitness in terms of how they how agents engage with each other and and how they leverage their own Fitness um intrinsically um so we talked we I think I provide a good overview of gold directed Behavior now what I'm going to talk about um so um that kind of so I'm going to talk a bit about um this this aspect of Regulation like trying provide more uh Clarity on like okay well the the uh as agents we're trying to regulate uh we're trying to to regulate an optimal means of sustaining an optimal level of fitness and and how does that work out right and so um maybe a number of you aren't familiar with David Walbert but uh David walber um talks about uh This this term known as this um this idea of guided random search optimization so I'm gonna say that say that again it's a an anandra of words guided a guided random search Behavior which has to do um so if we think of Randomness we're thinking about like this this act this uh this uh um uniformity in which uh uniformity in which certain outcomes occur certain things can happen right like that there's no uh there's no uh there's no uh prioritized there's no there's no prioritizing of of of certain um ways things might occur but there's uniformity in which occurrences happen and we could and so this uniformity um especially understood in in in in literature regarding prob probability Theory and uh denotes to what Randomness means not it's not necessarily like a lot like some people might think Randomness has to do with spontaneity and what have you and that might have to deal with probably extremities of of of Randomness per se that's not necessarily random um but uh Randomness more so as to do with with uniformity that there's no there's no like selective favoritism about how outcomes happen but that you know there's there there's there's kind of like this evening out of like how outcomes and circumstances uh happen in terms of thinking about what Randomness um implies here and um David Walbert approached this aspect of of thinking about um how we um develop axiomatic systems and um how we can think about um you know when we think about axioms we think about really us you know how we how how we how um how behaviors you know how can how things are reduced um you know how far can we reduce a a certain Behavior or phenomena and and when there are certain components that can't be furthermore reduced but can can you can use those components as building blocks to represent uh complications or complexities of how Behavior Uh how behavior Can can happen with a system now now we're thinking about axioms right so um uh a good example of of of thinking about ax is thinking about how arithmetic happens right like how how we can add or multiply numbers to get more numbers and we get AR we have we have inputs of numbers and we then we have an output of numbers we have a range of numbers so now we can think about when we're when we're talking about agential systems and we're thinking about axiomatic properties of systems we're really thinking about um you know how do we reduce stimuli and Sensations and generaliz the way the way Ag gental Systems generalize how to engage with stimuli and um now um I I just realized that I probably didn't give a give the give give the give give a give a an elaboration on actions and priors as opposed to perceptions regarding the P Cycle so let me do that real quick before I go go more into what is guided random search behavior um which then leads to this aspect of guided random search optimization because ultimately uh we as agal systems want to optimize Fitness right um now okay so perception has to do with uh perceiving uh you know we're we you know as we're engaging with stimulus and processing Sensations we are for you know we're perceiving uh we we are trying to figure out okay um how do I engage with this you know what what you know in terms of top down you know we're talking about top down here when when we're talking about perception actions and priors um although uh more simple agents uh it could be argued that you know even simpler agents have a means in which they perceive act and anticipate even though you know that their capacity the capacity more simple agents to engage with the environment is is is is less complex because uh you know if we're thinking about especially single cell systems as compared to you know humans we're these multi-agent systems because we're we're composed of multiple cells and um we have these uh networks of neurons and go cells that are facilitating engagement and and communication with different regions of other neural neural and gu cellular networks that are facilitating our capacity to perceive act and anticipate so action has to do with how we're responding to the stimulus right so we're trying to facilitate a response uh to to to the stim to to to how we're engaging with our environment we're talking about action and then prior or anticipation has to do with uh ke like a history you know we keep track of a of a history of our perceptions and our actions and we then we try to optimize based on that history we were trying to anticipate uh what H what what what perceptions and actions ought to be optimized going forward so on the one hand with perceptions uh uh you know we're we're we're you know we're we're engaging with an environment we're engaging with stimulus when we're trying to figure out okay how do I react to this how do I respond to this and then next in line once we once we once we have a calculation in mind of like okay this is going to be my action this is going to be my response to the stimulus and now when we're thinking about perceptions actions and priers um we again we we can think about compositions thereof of of perceptions actions and priors right you know we we think about not only U you know even even different modes of perceiving and acting and and at times uh they you know perception and actions are sort of coupled you know based on like how different regions of the brain are are processing Sensations and and you know once you process Sensations you you have information to work with and then you're prioritizing what information matters after you prioritize what Sensations matter um now okay this guy did random search now we're going back to to axiomatic systems um right we have this red we have we have a we we we get to this point where we try to reduce uh Sensations right we try to to decompose you know like especially when we're processing Sensations and and we're trying to um conduct this capacity to perceive act and anticipate ultimately we're we're trying to come up with ways to to you know as we're coming up with ways to predict we're trying to uh you know we're trying you know thinking about like that example of arithmetic we're trying to to you know reduce what the sensation to what we try to reduce to uh uh you know like Sensations only reduce so much yet uh compositions of those Sensations could start to build a picture right like the bare reductions of those Sensations might not make much sense themselves but if we take the compositions of those Sensations over time and how they build up uh into something that makes sense for us to you know in terms of interpreting the stimuli in terms of perceiving the stimuli and then acting on the stimuli after uh you know facilitating uh okay how you know in the p in in the perceiving of the stimuli what's the best what's the most optimal manner I can act on the stimuli and then having a history of okay well I in I perceived this stimuli this way and I acted on it this way this is what happened okay well the outcome might have been good or might have not been good so now I'm going to anticipate how to perceive and act based on the the outcome that may or may have not been good and um in terms of of of how my capacity to evaluate weight as you know you know when we talk about uh how um let's see um my capacity to evaluate the the stimuli is reflective of of like okay well uh I don't want to call it trial and error here but this is kind of how we're getting to the guided random search aspect right that there are there there's a combination of ways that we take um that we take the reductions of the sensations that build into information right and then we that become information this like this this reduction of generalizations of Sensations that become information uh based on how we're prioritizing that due to how we um select out perceptions and actions and PRI that optimize our fitness based on how we're engaging with our environment um so like um the guided aspect is uh prioritizing what's working as opposed to what's not working and the random aspect has to do with okay well the sensations as we you know we reduce the in when we reduce the amalgamation of Sensations that ultimately like I mentioned are stimuli that we're being bombarded with um though the aspects of those Sensations really are uniform um uh in in the sense that um those you know when we don't really prioritize those sens you know once like uh before those Sensations become develop any relevancy as information uh there's no you know the environment doesn't really necessarily dictate what sensation what stimuli have priority the stimuli just present themselves as stimuli um to to to to our system and then it's up to us based on how we perceive and act and anticipate the stimuli on how to how to it's up to us to prioritize um that that stimuli which which in uh in the Bare Bones aspects of that stimuli outside of our judgment you know outside of our own judgment which involves perceiving acting and anticipating is random because nature doesn't really care you know um nature doesn't really care about you know reg you know necessarily selectively regulating the stimulus outside of the agents uh like like the environment the bare environment doesn't care but the agents among that environment you know are going to engage with the environment in a way that prioritizes stimulus and and and guides the agents to engage with each other in the environment right and and in that sense the guided the guided aspect is the prioritization the random aspect is this uniformity that the the bare bone stimulus presented in the environment among agents there's no there's you know until the agents start to prioritize the sensations as information um there's really there's really um Beyond there's really nothing more than this uniformity of this you know there's nothing there's nothing really uh that that stands out about the stimuli until the agent uh starts to uh make selective uh uh generalizations once once there once the agent can notice patterns of stimulus and those patterns would be information and then the agent's going to try to Prior prioritize those patterns so the search Behavior now the search behavior is recognizing the patterns and then fil the filtering aspect of the patterns right so like I mentioned the the guided aspect has to do with the prioritization the random aspect has to do with okay well there's this uniformity in which the stimulus is is is provoking the agent to engage with the stimulus and then it's up to the agent to kind of categorize and and and and prioritize stimulus and and reduce it to information and then prioritize that information based on like what the agent think thinks is necessary to pay attention to and so the the search aspect is like this paying attent like the intentional Dynamics involved with prioritization um so um now there's like I mentioned there's constraints within the environment that enable agents to to facilitate uh perception action and prior uh Dynamics um in a way that would optimize Fitness as opposed to um deceive Fitness or or be counterintuitive to Fitness and now okay well when we talk homeostasis uh again the environment's not in a fixed uh static State Dynamic uh agents aren't in a fixed uh fixed state right everything you know we have this complex adaptive system that is the environment and we also have this complex you know the nature of agenes of Agents uh invoking agency in the environment or or trying to compensate for what the environment presents to agents as resources or obstacles um homeostasis has to do with then um how do agents sustain a minimum viable U functional Integrity that is that is that is crucial to Fitness to allow the agent to continue to continue to exist rather than than cease to exist right so the so homeo so homeostatic regulation um is going to have to do with um and maybe some of you are thinking of the free energy principle um in this respect that um agents um want to minimize cost but maximize on opportunity to uh leverage optimizing Fitness um what least most agents want to do that um very it's um I mean there there are some aspects where where I think we're agents that have the capacity to annihilate themselves um which um I I don't think uh I think agents need a certain level of complexity to make that decision from the top down so so so in that respect the top down Dynamics as opposed to the bottom up Dynamics uh you know are are are much more crucial to facilitating the Integrity of an agent because you know agents with enough sophisticated agency like human beings can decide to annihilate themselves which is unfortunate but uh in most cases it well I would say it's generally unusual for for and we're seeing this in the literature too that it's unusual but it's interesting um to uh you know where circumstances arise where an agent makes the evaluation that they can't really optimize Fitness anymore than they have so they so the most optimal the most optimal decision for the agent is decide is to decide for them to not EX exist anymore um which is unfortunate um but um this is not Al this also isn't uniquely human either um you know other other other there are other types of Agents other animals out there you know when we think again we're thinking about agents as biological organisms here um you know where an agent makes a decision that that they can't really optimize Fitness anymore and the the uh the the circumstances in which the environment provides any leverage of Integrity is not really there so the most optimal decision an agent can make is is to annihilate itself but that that's I'm not going to get really into that um but I think it that's sort of an unfortunate circumstance for an agent to find themselves in and and this pertains to probably um I you know more sophisticated agents um as as I found in in in the literature okay see so we we went over that we went over what stimulus is and regarding again homeostasis that has to you know this Builds on guided random search and the guided this being the prioritization the randomness being the uniformity in which stimulus provokes the agent and presents itself within the environment and we acknowledge that stimulus is is really um anything that in that that is provoking the agent to perceive act or anticipate anything that anything that H that en that that provokes an agent to perceive act or anticipate would be considered a stimulus um and prioritizing where was I here being able to to prioritize stimulus and for agents to minimize cost to engage with their environment uh and maintain um um like a minimal F minimum VI minimally viable functional Integrity that that is crucial to Fitness this has to deal with homeostatic regulation or homeostasis this is what I call homeostasis and the environment presents means in which uh on the one hand can be resourceful for agents to to regulate their Fitness while on the other hand the the environment might also present obstacles that could be problematic to regulating Fitness so we could so okay we're going to talk more about the mechan like the mechanistic sort of a mechanistic interpret interpretability um explanation of of the perception action and and prior cycle or what I call the pap cycle so uh a lot so a lot of you programmer types and Engineering types are probably familiar or hopefully are familiar with the terms sensor and actuator so uh on the one so we could reduce all the all these behaviors of agents perceiving and acting on the environment as as well as anticipating to be reduced to sensory and actuator based Dynamics sensory sensors being to sensory mechanisms like recognizing and and and um communicating or signal you know recognizing and signaling you know when we think about how how neurons neuron and G's uh cellular networks work together the sensor capacities have to do with um interpreting a signal and passing that signal along to the right Networks then deal with that signal and then you have this coupled capacity of sensory uh uh of sensory motor actions involved that that deal with the stimulus so actuators have to do with acting so so so like having an action towards the towards the stimulus so actuators uh uh having this actuation based mechanism um has has to do with responding to the stimulus while the sensors or or the sensory capacities signaling uh you know the signaling and communication faculties that opt that enable uh perceptions and actions and anticipation Dynamics to be optimized um you know certain networks of neural uh um there are certain uh there are neural and gal cellular networks that that uh interpret uh Sensations and communicate uh what those those Sensations are uh you know and they signal to the to networks that deal with the with acting and responding to the stimulus right and and more so there's uh it more there's this coupled capacity of you have these coupled sensor actuator based networks that um and compositions thereof that deal with handling stimulus and Sensations and prioritizing the sensations into information and so on and so forth and and the sensor and actuator based mechanistic capacities you know you know that that we reduce this you you know we you know basically this is this is this is a reduction of like okay we have these neuro uh networks that facilitate perception action anticipation and we're trying to optimize uh for metabolic costs and facilitate cognitive load uh that um that leverages Fitness rather than compr compromises Fitness um over time um and and and um in in the most realist I guess in the most uh pragmatic sense we think about this minimum viable means in which uh we facil in in which an agent optimizes how it calibrates cost of metabolic load for homeo for in with regard to facilitating um functional Integrity or homeostatic regulation so hopefully that makes sense so we could think of like when I mentioned gold directed Behavior there's this coordination of micro and macro levels of how we as multi-agent systems facilitate that right so like we think from the bottom up we have metabolic uh load handling or or facilitation and from the top down we have cognitive load uh handling right we're thinking about how cognitive load is mitigated due to how metabolic load plays out as cognitive load really is a subset of of metabolic load because you need you need you need metab metabolic costs to facilitate everything that an agent does like without being able to uh metabolize the the agent just wouldn't exist it wouldn't be able to continuously engage with with its environment because it needs resources like the agent needs uh fundamentally one of one of the fundamental goal goals of the agent is to consume resources whether that that's food water you know we think about human behavior we think about food water sleep um that involves the consumption of metabolic load and then uh of course other you know you know other means like using tools that um extensively enable us to better leverage Fitness and how we engage within our environment and other agents is is also important but that's more more secondary uh in terms like um you know like tool use and and augmentation and development thereof um is not as crucial as as maintaining like uh metabolic load is in terms of of of of food you know consumption of of food and water being humans like to to regulate that fundamentally so we can think of like on the macro level um cells represent you know as multi-agent systems on the macro level you know cells can represent or whole organ systems and subsystems that regulate from the bottom up as well as the top down and then from the micro level we think about how the individual cells as individual unit agents engage with each other because we ourselves even though we are agents we are a whole environment and ecosystem thereof of Agents communicating with each other that enables our own agency and our own capacity to communicate the environment in which we're embedded and so we have agents embedded within us and then we are also embedded within an environment in which we have ageny and that is constrained due to um the resources and obstacles that the the environment presents uh or the that that ex the resources and obstacles that are existent within the environment that that we have constrained access to and then the um there's also the um this is not talked about that much in the liter but there's also the Meso Level so we talk about the micro level where we have the individual unit agents within our organism as a multi-agent system how how those agents communicate with each other but on the macro level we have the different subsystems communicating with each other and even on the macro level we're thinking about the top- down dynamics of regulating executive function which also considers the macro level about how networks of of of neuron Gile cells are communic ating with each other but we also have this Meso Level and um this Meso Level we can think about how selective uh more so when we're thinking about the top down regulation of agency we're thinking about how different Sub sub regions of our of our nervous system or our executive functioning are regulating agency our capacity to continuously engage with our environment in an optimal fashion so um so so thinking from the Meso Level you think we can think about how different regions regions of the brain are are working together to evaluate perceptions actions and priors like you know like the prefrontal cortex uh the reticular formation the the hippocampus the amigdala like how are all these subsystems of of neural and gal cellular networks facilitating engagement um like we're going from the more macro level of the of the central nervous system to not necessarily the micro level of the individual neurons and and Gile cells but we're also talking about how different how uh different coupled networks uh of these neuron cells of these different regions are facilitating lateral engagement with each other so I think it's important also so for those of you who don't necessarily have a neuroscience background um or psychology background um uh so no no one region of the like one like all the regions of the brain first of all are are communicating laterally in the sense that they're communicating they're they're engaging in parallel processing stimuli and Sensations and what have you and and information right no one region is responsible like you know the hippocampus uh doesn't doesn't store and and you know even though like the hippocampus is largely recognized as the memory storage system the way it stores and and references information it needs to work with the amydala the phus it has to work with the reticular formation um it has to work with the prefrontal cortex um um so so no one region is processing these Sensations at one point or momentarily in time like they're you know and we're in in an interesting way we're going to get get to like um what is in necessarily interesting about um how our brains are structured and that's not necessarily uh a het it's not necessarily hierarchical although it's it's mostly hierarchical in the sense of prioritizing stimulus and sensory information but um how uh how how that processing and prioritizing builds out over time in terms of how we as organ as these biological organisms as multi as agential multi-agent systems evolved um is not just hierarchical but uh is there's this other term that is less familiar um so on on the on the one hand we have this prioritization or or what can be deemed as hierarchy of like o of of prioritizing uh of Sensations and and and what have you on the other hand there's the tuning aspect right like sort of like as we're prioritizing and selecting out uh stimuli loads as sens and as Sensations and information uh there's also uh this aspect of sort of like tuning you know like you know kind of like you know if you think about like a like an audio mixer or or a soundboard right you you want to get the right you you kind of want to you want to kind of Select out the right uh channels of how stimuli and Sensations are being prioritized and this is known heterarchical processing h t uh heterarchical h t archical so I'm not going to spell the whole thing out but uh this is heterarchical processing and and this term is actually brought out in um uh uh H H hubner and schulin biological cognition paper which I highly recommend uh and we're going to talk a little bit about um this uh this this u means of preserving viability which um which is pretty important to how agents engage with each other to optimize and leverage Fitness we're going to talk about that momentarily um but so there's this there's this hierarchical capacity in which the regions of our brain are prioritizing stimulus right and and and um in that prioritization and selecting out the stimulus there's sort there's this tuning coming there's there's kind of like this channel tuning uh mechanism right like trying to just get the you know trying to trying to prioritize the stimulus just the right way involves like sort of like a selective pressuring that's not necessarily hierarchical but it's it's sort it's uh um it it it's also like um a little bit more it's sort of like there's degrees of freedom in which we we we try to calibrate the channels in which those Sensations uh are are being prioritized as information and then then that information is being selectively prioritize uh so this is known as heterarchical processing um but largely speaking what's happening is hierarchical because that the actual prioritization of modalities that's hierarchical but that can't happen without calibrating uh the sensations and information to be processed and this calibrating aspect is heterarchical um or like kind of like you know think about a mixing board or a sound board like like I don't know how many people are are are into DJing or or or producing music or what have you but if you play with like a soundboard and you're trying to get the right sound to come out and you're and you're trying to get the channel properties you're trying to tune the channels and all that stuff uh this is this is this is hyp you hyper parameter hyper param it's a tongue twister hyper parameterized tuning processing of the top down right like the idea of parameterizing the sensations right like when we're trying to facilitate the parameterization of okay the stimuli coming in and are perceiving and acting on that stimuli um the the param par par parametric tuning that that has to facilitate the prioritization that is the heterarchical processing that's happening okay so so there's there's um on the one hand there's what what what's happening with perception actions and prior and priers is this introspective capacity of of interpret uh stimuli and and then this reactive capacity uh this proprioceptive capacity of aligning our perceptions and actions uh with with how we anticipate right so so our so so in terms of not only our our um well this is in due part to to the top down functioning but this and this this really calibrates uh our capacity to to not only uh interpret but react and and and and and react in a sense uh react to stimuli uh in a sense of like uh uh of predicting what's going to happen what what's going to happen going forward uh that would be Pro more more in the proprioceptive sense of dealing with the stimulus over time to optimize Fitness so on the one hand there there's this capacity in which uh um especially with decision making where uh we try and make sense of what we ought to do how we how we how we perceive and act on the stimulus then our continued actions going forward in time has to do with proprioceptive capacities um so so this capacity to to the means in which to to to to to to regulate perception and actions based on anticipations um going into the future and and how how how in terms of bodily functioning like just just how I engage with the stimulus in in a way that becomes more that feels more intuitive um this is appropriate receptive capacity as opposed to more introspective capacity of uh uh of of in trying to interpret and understand the stimulus and and in in and making a decision that might be and making decisions that might reflect um uh what could happen in the future then there is The Happening of the future right and and the alignment of that happening in the future that that that it that it that um that is in that is in line with anticipation has to do with proprioception okay let's see um I don't want to speak too conf confidently about about the main regions of the brain coordinating uh coordinating our capacity of perception actions and priors but I I'll I'll I I'll I'll bring some some key some regions that have caught my eye um there's a bit of contention in terms of what are the key regions that are are responsible for our capacity to perceive act and anticipate uh the most evolved capacity uh that really sets humans apart uh has to do with the prefrontal cortex so that that deals a lot with planning and and and perception that there a lot of the modalities in which planning in terms of perception and gold directedness has to do with the cortex um the anterior CeX has more to do with the sensory motor functioning um now um less what's less talked about in the literature um is the role of the reticular formation so that's your brain stem so the reticular formation has to do with autonomic arousal so so this like the capacity to even uh uh uh engage with Sensations at all uh has to do with the reticular formation and in particular what I uh my favorite uh part of the brain um is the locus corus or also known as the the LC NE system or the the the what regulates norepinephrine um so so so autonomic arousal is pretty important because what without it like we're we're we're not really engaging you know um I mean there's no means for stimulus to to to be interpreted and and facilitated throughout the uh the brain so so so the norepinephrine system is is allowing us to interpret this the stimulus and guide that interpretation in terms of signaling uh all these all the different primary primary regions of the brain uh that are responsible for perception action and anticipation and now what I like to call The neuromorphic Interchange is the limic system now the lyic system is basically uh basically the region that encompasses like basic uh the the key regions of executive functioning so thinking about memory retrieval motions um plan angle directedness um sensory motor processing so the prefrontal cortex at least most most of the prefrontal cortex uh or the Neo cortex layer uh the hippocampus the amydala the you know all all those all those really important regions that are are are important for are for executive functioning are Guided by the the limic system right uh well they're connected together by the limic system like they they facilitate each other's capacity due to um being housed into this interchange that facilitates their engagement which I call the limic system and uh let see all right going to the next section Sensations are fundamental to perception action and anticipation I think I covered most of this so this idea of generalizations of Sensations um is this feeding of stimulus be uh as inputs into into multiple our multimodal way of of engaging with the stimulus uh has to do with this General of Sensations in which that the stimulus is fed into the system initially and this is this is a this is the visceral uh this is this is the visceral of the stimul the viscerality I guess viscerality is the right word so the means of internalizing the stimulus is this generalization of the stimulus itself right which is visceral in natural right like think about like when we think about visceral like think about how blood flow is just naturally happening like throughout your body or or certain like like even metabolic uh metabolic processing um when we think about uh uh uh the um glycolysis and and all these different behaviors that happen that that regulate metabolic processing uh cellular respiration right all like stuff that's just constantly happening that that that regulates metabolic load uh this like that's generally Prett you know that that that um like like these behaviors that that aren't really particularly that interesting that that correlate uh directly to to our executive top down functioning this can be recognized as visceral processing like uh you know uh U like stimulus that that isn't that that's just constantly happening or or processing this this load processing that's constantly happening that that's not that interesting but it it's it's kind of fundamental to like how how our continuous engagement with the environment as as as agential systems is regulated that's what we can denote as um visceral processing and this internalization of of of stimulus uh with respect to environmental constraints of course um has to do with um you know we're we're filtering the noise in this generalization of Sensations like this this capacity of General uh of filtering the noise into becoming um a means in which we can make sense of it um so we prioritize that noises information and that eventually allows us to anticipate you know how should we perceive and act accordingly to the stimulus being presented to us and then we have these recollection of experiences because we're able to anticipate based off of our history of perceiving and acting to stimuli now now here's where things get interesting because I'm about to talk about system one and system two uh coupled with the Rider and the Elephant uh so uh system one and system two uh refers to Daniel conoman uh in uh where system one has to uh uh decisions and be and ways of act engaging with our environment uh from a top down level executive functioning level which is which is more automatic and less involved while system two has to do with um non-trivial or problem solving circumstances which take more effort so now that has to do with with with reasoning and and how we optimize engagement with our environment uh now um we couple this Capac with Jonathan hates the Rider and the elephant and the reason why uh this isn't this is well this is this is of interest is um especially with system 2 processing we use emotional veilance processing um or emotions these these feelings the feelings we get from from the general generalization of the sensations that we we have as these embod um uh as these embodied beings but like as AG gental Systems our engagement is embodied right this this this capacity to generalize Sensations um as agents is is embodied which this is what really separates us from from classical machines or just just you know these closed loop circuits um and I would argue additionally that um the way that we perceive act and anticipate is not necessarily A a closed loop um it uh it it's sort it's sort of like an open system but it's not uh it's like a bounded open system um we we'll we'll get into that a little bit soon um now um what's interesting about uh so talking about the rider and yelf and I need to elaborate on that so system one system two system one is like kind of automatic doesn't take too much effort to make a decision and and and and and system to it takes a little bit of effort it's non-trivial uh now the Rider and the Elephant the rider is like the voice of reason while the elephant is like the experiencing like of the of the the feeling States right so one can only imagine an elephant going rampant in the village right like the rider doesn't have a good grip of the of the elant the elephant just go crazy um so I so I couple this idea of the Rider and the elephant with system one system two because um especially with system two it takes some it takes some uh you need a way you need a a a non-classical logical uh system reduction way to uh take uh uh feeling States emotion States in in where uh decisions are not are not trivial or or they're not like there these non there's these nonlinear non-trivial circumstances in which we have to make decisions or make predictions or or you know like how we engage with stimulus and how we try to make make anticipations of how should we perceive an act going forward into especially the distant future right that it could be a big deal in terms of leveraging Fitness so um so so this uh so so we so what we so what happens here where decision making is not so intuitive is we can try and figure out a way to have decision weights as criteria to evaluate decisions where there there is some uncertainty involved where we're not certain how things are going to pan out and we have to anticipate as agents like how how should we engage optimally with our environment emotions uh play a role uh in this capacity where where decision- making is non-trivial nonlinear um and um there's this generalization of feeling States right when we when we're talking about generalization of Sensations uh so first we have to consider that raw Sensations have to be process by specialized regions which of of our executive functioning uh that deal with memory consolidation uh regarding optimizing priors as well as um emotional veilance processing so effective processing uh is involved here um so since so what happens is Sensations become these streams of information that are comprehensible for modes of reasoning and and emotion balance waiting that can optim more optimally guide perceptions and actions on PRS um now um what can be understood as emotions uh then would be generaliz this this veence weight this emotional veillance waiting on generalized feeling states that um is is separate but necessarily uh is separate but necessary for optimizing rationale and according accordance to decision-making so we have this interwovenness of reasoning and emotions or or feelings right espe and and emotions kind of um uh provide a a means to weigh decisions that are especially more difficult where it's not obvious uh what we how we should anticipate uh what's going to happen in the future and and just how to guide uh our our Behavior Uh especially in moments of uncertainty um and uh it could be argued that consolidation of mem memories would be largely compromised if emotions did not provide emphasis on how memories are referenced in the first place because you need a reference pointer and that's what the these emotions themselves like these emotion weights themselves function as like a a reference a point of reference uh in how we rationalize our our our be our our perceptions and actions uh based on how we anticipate so in accordance to that if we can't uh if memories cannot be properly referenced on weigh decisions then how how how can our how can we optimize perception and action according to minimizing uh compromising Fitness right uh that that would that's a that's quite an existential question like if we didn't we weren't capable of emotions on some level like how would we carry Baseline decisions that allow us to effectively anticipate what what do we do next B based on our environmental circumstances right um maybe that's not super intuitive but to me that that kind of is um so we talked bit we talked about system one system two and the Rider and the Elephant we talked a bit about hierarchical and process hierarchical and heterarchical uh processing so again hierarch has to do with prioritizing stimulus and information and heterarchy has to do with the like how we parameterize it how we select it out in in the in association with the with the prioritization so they're kind of coupled together but of course priority uh is what the priority is what matters even though we select out certain perceptions and actions in accordance to how we're optimizing our engagement and and hopefully I elaborated that clearly but um I could always go back to that a little bit later if there's questions about that uh let's see relativistic I'm going to skip section four regions all right so this is this is uh probably one of my favorite parts of the paper uh right here is talking about reality as an interface or or reality is an interface um and uh this is kind of this was this gained a lot of popularity in Don Hoffman's collaborative paper with um uh with Chris fields and chatan Pros um especially elaborating on like the boundary you know we're talking about perceiving and acting and how agents are you know agents have a boundary within the environment they're engaging with right but even more interesting is that I make the case 4 is that agents themselves are existentially embedded within the environment that they're engaged in and uh this can argue I would say arguably this could extend to artificial artificial agents that they rely on some on resource leverage within the environment that they're engaged in for them to continue to exist or or function right and uh now reality as an interface uh we think about this this uh you know this phenomenal capacity for us to engage with stimuli and and and and and react to it right um and and this can this can like very deeply uh you can really argue reality as an interface like deep down P you know past the fact of of organisms and talking about physics and and and just how SpaceTime uh that the the the the fabrication of

SpaceTime maintains itself is that uh particles are interacting with each other in some capacity even though they don't necessarily have any agency uh but most importantly we as agents are constantly engaging with each other we're constantly interacting with each other so to speak in the phenomenal sense so that really has to to deal with reality as an interface and extensive and building on that we talk I I mentioned axiomatic systems uh sort of as the reduction uh you know we we we're we're we decompose Sensations uh reductively axiomatically and and we can relate that to David walbert's argument uh of talking about uh okay well we we base a lot of our we base our reasoning on mathematics mathematics is based on axiomatic uh constructions of systems um and we have base components that allow us to uh construct Pro uh to elaborate on properties of of systems or construct systems uh compositionally in a number of ways uh based on uh like like base base uh like Baseline uh rules like for instance considering the rules of arithmetic right um let's see is there certain things that I should we talked a bit about guided random search um so this idea of abstract reasoning is rooted axiomatically because making decisions involves uh involves again trying to make sense of the sensations that we're trying to generalize and we're trying to prioritize those Sensations as information and then we try and prioritize that information um to make sense of the world around us in which we're constantly interfacing and engaging with so um now we uh come to an interesting dilemma of uh when we deal with this understanding of of okay well we're constantly engaging with the world around us and we can decompose uh our engagement um axiomatically so to speak especially when we uh try to reduce Sensations and um we we try and figure figure out how to filter those Sensations into information and prioritize that information based on how we prioritize Sensations as information then we're prioritizing the information um okay well in mathematics there's uh this this this this hole um that is brought about uh due to Kurt girdles incomplete theorems um not really going to get in that into that but uh what really should be um what what what should be uh the moral of the story is uh that uh the lack of consistency um in how we construct you know in the sense of how how gerle uses uh uh uh uses uh the ax uses the axioms of arithmetic as an example and trying to uh map those axioms to how we construct the natural numbers um there's uh there's inconsistencies uh that he presents um in that um and I guess this this more so uh in you know this sort of Pro provides an insight into uh the inconsistency in how we generalize Sensations as information and then we try and generalize information and try and have a consistent interpretation of reality which um in the illusionists which in The Illusionist sense uh you could argue that we never have a true represent ation of reality but um enough of us can come to an agreement on some representation on of reality based on on how we formally rep uh uh how we formally rep uh uh represent uh how we formally present models of the world around us so in terms of making predictions about how certain of aspects of reality Works how certain aspects of of the environment like how how how do we generalize uh ways in which we can engage with the environment not for ourselves but to facilitate cooperation with other agents right like if there's there are means in which we can collectively come to some agreement that makes sense in the moment and then there's a better way to make sense of it in the future uh more so The Illusionist aspect is that we're not going to you know we would be deluding ourselves elves if we truly thought like understood reality for what it is rather than really uh generalizing uh certain features of it that uh pertain to optimizing Fitness and our survivability and our engagement so ultimately we're we're exploring our environment and we're and we're just trying to uh optimize our Fitness with respect to how we engage with each other and and ourselves within um and uh the inconsistence the inconsistency is is the general the generalizations that we make that um become replaced with better models in the future so that's not necessarily uh that's not necessarily uh the same exact reasoning that girdle that Kurt girdle makes saying well we have this inconsistency of mapping the uh properties of of arithmetic in which we can we can compose certain uh systems with arithmetic uh and map that to natural numbers in which we for which we assign each composition that we come up with using arithmetic uh as uh some um iteration within the natural numbers and then we come to some property that says that uh this property disproves that this prop um this property shows that um this isn't true uh well uh yeah you you come to you come to a property uh that that that shows inconsistency say saying okay well uh this this isn't true uh how did it go yeah basically you bump into to to a problem where you eventually uh formalize a a condition that is a contradiction saying that uh uh that uh that that that is that uh that it's escaping me right now but essentially you you bump into a formalization of a contradiction within trying to uh compose conditions that are consistent with with the axum aums of of arithmetic but anyway that's kind of outside of the that that's outside of the scope of this presentation the the whole means of like the whole point about inconsistency is that we're constantly generalizing how we're engaging with the world around us uh as agents uh okay you're right uh so like basically what girl came up with a with a property with with with with with a condition that said uh like this this card proves that uh that that this that that you know like this this condition proves that it's false otherwise it's true right so like you have a condition that proves that something is not untrue but it it it but it's also true that it's not true that that's the contradiction in a nutshell um that's not exactly how he phrased it but I'm like talking off the top of my head um but you basically arrive at a contradiction um and and kind of sort of uh relating that back to how we generalize Sensations and information is we never have a completely consistent picture of reality um especially how we engage with it and we're constantly reform re reformulating how we interpret our reality for what it is and and that itself is the illusion not that re reality isn't an illusion but our understanding of reality is elusive because we're constantly generalizing um what reality really means I think we're ready to talk about preservation of viability okay so this is where uh and I spend a lot of time in this paper talking about interagent Dynamics in terms of like how cooperation is essential for how we uh as as how we even got to Modern civilization as as humans like if we didn't cooperate well enough together we just wouldn't have gotten you know we wouldn't have zoom calls we wouldn't have gotten to where we are if there was not some sufficient means in which we we cooperated with each other effectively as sufficiently large groups of humans um and this goes back to this preservation of viability which really has to do uh with how um we consider tradeoffs in engaging with our environment while also we consider how uh we engage with each other and recognize and have this awareness uh of our own degrees of autonomy with respect to other other humans as agents but uh we aren't the only agents that exists within the environment there are other agents that aren't humans that don't have the degrees of autonomy that humans have and we ought to acknowledge their existence as well and try to acknowledge the tradeoffs in optimizing how we engage with the environment and other agents that aren't humans uh that I kind of call lesser agents um that they don't necessarily have the the the same degree of autonomy to engage uh with the environment around them and I'm going to get into that in a moment um but really there's this there's this idea of this uh optimal e and flow of guiding our priors with which is not limit to limited to individual Fitness of us individually but also how we engage with each other so there's this there's this idea of interagent coordination uh that ought to be facilitated and and we could think about this in a diplomatic sense of like how we engage with each other um and and ultimately this increases the well-being uh of of each other and ourselves and our overall odds of survival and acknowledging that that you know uh the Dynamics of modern society itself uh where we fluidly communicate with each other through evolution of language technology and bartering would not be possible if we did not rely so much on how we coordinate and cooperate with each other is that that's critical to just how we got to where we are as human civilization um and engagement uh with each other not only increases our chances to optimize Fitness but it enables uh our capacity to further develop uh our awareness and means of engaging with our environment in a way that can become more optimized over time at scale right because more and more of us exists we have to think about scaling uh problems with scaling and and how we collectively engage with each other and consider perception and actions of others not just our own uh perception and actions and how we anticipate but as as collectively how how we how we all coordinate and cooperate in such a in such a fashion to to to facilitate that um so so this is there's this idea of co-inhabitants and it's not just humans that exist on this planet that but there's other organisms that uh do not have the degree of autonomy that we of agency we have but they serve functional role within the ecosystem for us to exist so I'm talking about all kinds of life like plant life uh you know uh not just animals but plants fungi molds like uh insects if it wasn't for you know the existence of pollinators and and all these different species of organisms if they didn't exist we wouldn't exist so we have to take we have to acknowledge that um you know it it you know this that we co-inhabit um and actually some literature that came out recently come comes up with this term Co co-evolution that we co-evolved with the with I don't use the term biosphere uh in my work other than the abstract but we co-evolved with all these species that we co-inhabit the environment with and we're all embedded uh we're all essentially embedded within this this ecosystem that co-inhabit and so I think there should be more awareness brought to that front in terms with how you know there there's so much there's such an issue with human conflict and just you know what I see as like this this uh uh this bias of not really recognizing at scale that we ought to do more to to to work together and symbiotically coexist with our environment uh we we ought to hold ourselves more accountable as a species because we have this we have this level we have such degrees of agency in a awareness that uh that gives us the that allows us enables the potential for how we can cooperate and where that can lead us uh I mean it got modern it got humans to Modern civilization which I think is absolutely phenomenal um and we're going to get a little bit into that um but um before I get into that I want to talk a little bit about uh uncertainty um so this kind of goes back to uh leverage and guided search um where um there are moments with regarding to calibrating our perceptions with regard to next actions that we take in accordance to the Past and Present Moment uh what could unravel in the future regarding optimizing our priors especially when we're facing moments of uncertainty right so um this ha this this has to do with complications that happen with regulating uh system 2 Dynamics um and uh this uh when regards to non-triviality nonlinear evaluations decisions um and uh this can complicate how we reason about the world around us and regulate our emotions and um uh and how we can persistently act effectively in a re in a way that's reinforcing our fitness so a persistent uh that uh persistent uncertainty with respect to conducting the capacity to perceive and and act effectively can hinder the efficiency in leveraging and reinforcement learning Behavior making us more immediately prone to acting maladaptively eror prone signaling of sterilizations of Sensations becomes more prominent in the face of uncertainty uh which can end up deceiving our our senses instead of being able to effectively optimize our priors in the face of circumstance so on the one hand uh dealing with uncertainty uh involves novelty exploration where we try to improvise and figure out ways to uh to how we can calibrate our perceptions and our actions with respect to optimizing our priors uh and then and and that has to do with with how we recall like how how we we're engaging with our environment in the past right we talk a lot about like anticipation with respect to perceiving and acting and and uh novelty really um you know it's not just the improvisation it's kind of being clever and um trying to experiment with ways to optimize Fitness in the face of um and this can be this can be leveraged in the short term as well as the longterm um where uh novelty can be leveraged by uh considering how we recall experiences in dealing with circumstances that are either out of the ordinary or just by happen stance making a recollection from some point in the distant or not so distant past relative to perception that may Aid in improvising against optimizing priors and uh we can consider this uh uh an aha moment like oh I figured out what to do um so in terms of novelty exploration and you know finding yourself dealing with uncertainty um that that's one aspect of handling uncertainty another aspect of uncertainty is just being is unfamiliarity um like an agent being unfamiliar with circumstances uh they're going just going to be have this uh sense of unfamiliarity of what to do and maybe other agents could help facilitate that that that have experience in dealing with those circumstances that that in accordance can help you know with cooperation can help the agent optimize uh those moments of uncertainty by becoming familiar or at least developing that familiarity themselves uh which might be prompted by novelty exploration um now now here so there's there's there's a couple uh there's two primary modes uh there's two there's uh how do I say this see okay so on the one hand um there can be a means in which uh there's problems interpreting uh what we can do from perceptive uh point of view um that deals with interpretation uh and this is this has to do with um now when we can problems with uncertainty this is sort of kind of like the halting I would I kind of call it the halting problem of agency where an agent doesn't quite know what to do until they figure it out or maybe they don't figure it out and you know that kind of Dooms the agent but um so on from an interpretive point of view um we can consider that uh there's problems with considering what decision to make in terms of uh uh the determinable SE the determinable sequence of actions in line with an agent capacity to perceive their surroundings considering contextual variation of intrinsic and extrinsic value um so that has to do with an interpretation being uh the problem of uh uh uh of uh aligning that that aligns with uncertainty that there's there the agent is having a problem interpreting what to do uh on the other hand uh there is this um so interpretation has to do with problems with the agent perceptively reflecting on their priors Moment by moment to understand what they ought to do to resolve problems with optimizing Fitness on the other hand there's problems with functional capacities of dealing with uncertainty where the actions determined where the actions determined and executed upon Moment by moment that fall in line with perception in alignment with priors um there that there's a problem with um there's a problem with actions being selectively and sequentially carried out with regard to context by moment so on the one hand uh there uh uh uncertainty could be a problem with inter with perception on the other hand uh uh uncertainty can also be a problem of functional capacities in terms of like not just perception but also like how the Asian ought to act uh with respect to the the circumstances they're dealing with uh and selectively carrying out certain sequences of actions Moment by moment where they're kind of stuck trying to trying to carry out uh the action as opposed to uh making a decision on on on on what how to interpret the circumstances they're in on the other hand um you know they also have problems with like acting on on the circumstances they're in as well uh what else should I is there something interesting to say on this so I'm not going to I mention that uh that this is like the halting problem in terms of uh um you know um having difficulty perceiving or acting uh for the agent that compromises their Fitness um the traditional sense of the halting problem is when uh there's problems where you condition an algorithm uh on certain inputs and and um you don't feed any information for the uh into a system for an AI algorithm to evaluate it so it just Waits uh for inputs to come in or or rather it just hangs there and it does and you know the system doesn't do anything it just is just waiting to engage uh with a user on this hand um uh the halting problem as I condone it to be is problem with dealing with uncertainty where the agent just has problems perceiving understanding uh you know know developing an understanding of of how to make sense of the circumstances they're in uh in an interpretive point of view on the other hand the agent also can have difficulty uh with under uh with uh action like like uh with with with with carrying out how they go about their circumstances right so hopefully that that makes sense comp complications regarding see right so I so not only uh is uncertainty uh deals with unfamiliarity where the agent is Con constrained to exploring novelty or novelty explor exploration but also um that U the agent might just not be familiar with what to do and they just need to figure out what to do and that not that might not necessarily be novel for all the agents that are that have dealt with that circumstance but it could be just novel for for the agent dealing with that circumstance and on on the one hand if they cooperate with another agent that understands what to do and that agent communicates what to do and that resolves that misunderstanding or um maybe sort of a novelty exploration sense the agent figures out themselves so I guess you can call it a novelty exploration case uh in that sense and um this this has to do with um uh you know these moments of uncertainty deal with error proness and compromise of system 2 Dynamics right so let's see and uh we and uncertainty itself uh really arises from moments of ambiguity where where information does not present enough Clarity for the agent uh to perceive and act effectively in alignment with priors to optimize um and I and I do kind of compare how that differs between computers like how comp you know comput you know Computing systems that we're familiar with is different um I'm not going to get too into that um so unlike uh let's see okay perceptions actions and alignment of priors become vulnerable to risk-taking Behavior with respect to facing ambiguity as behavioral Dynamics become a gamble on Fitness of the agent in question risk-taking behavior however is not necessarily maladaptive in its nature but is prone to compromise the agent's Fitness in which the the risks taken May likely become maladaptive uh given complications that happen with regulating rationale capacities and emotion through put under moments of ambiguity most especially for us uh human organisms become vulnerable to experience what is commonly known as anxiety uh so now I'm going to talk about anxiety I'm going talk little real quick about anxiety as a complication of ambiguity and um that really has to be that that's really worrying about what's going to happen in the future um largely um and being prepared to deal with those circumstances so when an agent is is in this dilemma of what it can do regarding future circumstances um then it's in in it's dealing with uh what I like to call anxiety um that would be some uh anxiety let's see here um anxiety Then would would be some level of Behavioral fixation which permeates due to lingering uncertainty in calibrating perceptions and conducting appropriate next actions this permeation ultimately presents self as layers of thought-based as well as compulsion based complexes that complicate perception with respect to generalizing next actions and aligning priors um given that anxiety is largely psychological the degrees of anxiety permeation are primarily due to nurture Dynamics where we consider constraints with regard to environmental exposure that can hinder adaptation most especially with regard to trauma response uh anx and so hypervigilance is also uh complicates anxiety or rather over sensitivity in processing stimuli as well as overreactivity to responding to stimuli which can be marginally rooted from physiological constraints hypervigilance can also arise due to behavioral fixations arising from complications and nurture Dynamics uh more concisely hypervigilance presents itself as complex behaviors that enable our human organism to act on alert with regard to optimizing our perceptions actions and aligning priors moments of hypervigilance can be useful when circumstances align well with Fitness however hypervigilance May pose to be maladaptive when it becomes compromised um most especially due to problems with upbringing or overexposure to certain environmental stimuli triggers so now we're going to kind of get into uh hyperarousal as opposed to hypoarousal so sensory processing complications that pose problems in calibrating perception are largely due to moments of noise filtering from generalizations of fil of Sensations being over undercompensated so this overcompensation has to do with hypervigilance while this under comp compensation has to do with hypo vigilance or or hyp hyper hyperarousal as opposed to hypoarousal and uh overcompensation of of sensory processing should however not be confused to be equivalent to some form of anxious Behavior but rather that moments of hyperarousal potentially complicates permeation the permeation of how anxiety takes form and persists and all right risk-taking Behavior with respect uncertainty so uh see so marginal risk is always associated with metabolic load demands and cognitive load demands um there's always some some risk associated with just engaging with with our with and uh you know trying to prioritize resources and dealing with obstacles um that that you know inevitably comes you know even risks that we're unaware of that we're unconscious of um and also there's you know just given problems with cellular inessence and aging and becoming more suboptimally fit in the over time and you know also presents risks uh inevitably uh so so here's a interesting uh bit I want to bring up regarding uh risk-taking there's a fine line between risk-taking Behavior associated with associated with uncertainty and risk-taking behavior due to intentional intentionally maladaptive Behavior given that we associate risk-taking with perceptions and action based gambling on Fitness cognitive biasing is involved with evaluating these gambl fundamentally speaking unless there is intention behind a particular agent deceiving itself we consider such Behavior to be risk associated with acting in accordance to uncertainty with respect to the agent in question clearly risk is associated with an agent's intentions be it self-deceptive or otherwise if there is marginal compromise to be considered in the case of optimizing Fitness in some fashion regardless risk is inevitable to the well-being of Agents as marginal compromise to Fitness not only is pertinent based on Executive functioning Dynamics in system one system to processes but even metabolic load processing Dynamics from bottom up that are not necessarily directly associated with executive functioning from the top down and this is the second to last section I have to go over uh reinforcement learning see what I can pick out of here this is really getting more really into reinforcement learning with regard to agency um so um deterrence uh is something I I realized didn't really talk about deterrence at least U the current mechanisms so this is interesting uh given the the whole point of reinforcement learning behavior is to leverage perceptions actions and prior to optimize Fitness uh a rather interesting expectation one ought to conjure is that reinforcement has much to do with deterrence from perceiving acting and anticipating maladaptively um deterrence however ultimately is not the goal of reinforcement learning based Behavior but rather some reinforcement learning based behaviors provide mechanism that mechanisms of deterrence but not all most especially if a habitual behavior that was once reinforcement based transitions into maladaptive Behavior due to environmental constraints impacting the Dynamics of an organ of an agine in question and such Behavior Uh cannot provide a mode of deterrence clearly on the other hand there can also exist reinforc learning based behaviors that act as modes of deterrence that become ineffective and thus transition to maladaptive Behavior being able to align perceptions and actions with regard to precautionary constraints that alert the agent in question with regard to priors effectively demonstrates a mode of deterrence okay so here's something interesting uh about deterrence um and then um I might go a little bit over language and how that's important to uh how uh we engage with each other uh in a Cooperative fashion that got got us as humans to where we are um but um so just want to close off the tance here um okay yeah I'm going to try and wrap up this this this whole section um right certain reinforcement learning based behaviors may not have precautionary constraints aligned to them although May momentarily leverage Fitness due to opportunities in which the agent in question does not incur Fitness penalties with regard to assess metabolic and functional risk that compromise overall Integrity additionally while agents may be aware of modes of deterrence they they may be intentionally oblivious to them with regard to agents being intentionally self-deceptive towards modes of deterrence the purpose of doing so involves Fitness trade-offs which in large part not only encourages marginal risk associated with Fitness penalties over time but also tends to be optimized with regard to anticipating a trade-off in which the agent in question is convinced will pay off in the short term or long term where things go AR with such self-deception and other means of self-deception varies with regard to the magnitude of maladaptive Behavior being enacted by the agent in question a rather paradoxical remark can be made regarding deterrence perhaps there are degrees of deterrence associated with reinforcement learning based behaviors as associated with fundamental uh the fundamental alignment of perceptions actions and priers such fundamental alignment is variable in terms of navigating our environment with respect to what we consider a fitness landscape of associated risks and precautionary constraints to consider these Associated risks and precautionary precautionary constraints enabled an agent in question to leverage optimizing Fitness and attentively Rec recognizing obstacles and aspects of the environment to be war of in order to furthermore avoid incurred Fitness penalties that could that would be otherwise obvious to the agent in question um so the rest of this section um talks about uh interagent cooperation based Dynamics uh and then the other section that I completed uh goes over Mal adaptive learning Behavior so um I talked a bit about uh cooperation of agents and how that's important so what separates us as humans uh from uh any other agents that exist is just our capacity to to utilize and develop language to a point where we can take information and and have what we could take that information and turn it into generalized knowledge us using language to uh we can take language and information and make associations that stay consistent over a prolonged period of time with how we as humans engage with each other and and are able to leverage uh the uh resources and constraints of our environment and and that impacts uh like how we develop social structures and and just this this this uh capacity of cohesiveness and and and how we utilize language to augment our capacities to better optimally engage with our environment and furthermore uh uh uh abstract the world around us and make uh generalizations of information and generalize knowledge um just get to it um the the the capacity to take an incorporated sophistic ated Lang structured language system and map that to Concepts uh using language um becomes generalized knowledge um and this is this is really what separates us from other agents and and how we got to where we are um and so we can think of like M medic adaptation capacities emphasized through Celia Hayes work on imitation learning and medic cognition Dynamics and more interestingly we can also uh consider uh France SWAT Chet's uh conceptualization of intelligence generalization on how we make generalizations about our perceptions and actions and and anticipate our environment but in so in in a collective fashion utilizing this capacity of structured of of of language and how we're able to like furthermore augment language as we utilize it and and and incorporate uh mappings to Concepts that become generalized knowledge because that knowledge enables us to to engage with each other in a way that optimizes technology and Tool use and our capacity to engage with each other and make abstract uh conceptualizations of the world around us and uh I could get really into that um um but I spent a lot of time writing about that but uh that that uh that that's really what makes us uniquely human now okay maladaptive learning now has to do with error prone habits and now getting um the real issue with maladaptive learning is what I like to call um uh well I don't know how many of you are familiar with an non V's work uh an non vadia uh brings up this interesting concept known as cognitive strain so we can think of like not just like you know if you think about physics and forces and how like applying stress like uh let's say I take a wall and I throw it at a wall and if I throw throw the ball at a wall at the wall enough times it creates a dent in the wall right so likewise cognitive strain uh we can consider that like the amplif you know uh cognitive strain could be uh the problematic Association of like what really makes maladaptive learning an issue um that um this prolonged exposure dist stressors that compromises Fitness that compromises the capacity for agents to have a a a consistent World model and to um leverage their capacity to to perceive act and anticipate in a way that's intuitive and and unfortunately uh like excessive cognitive overload uh compromises this so um this goes into like you know uh you know okay how do we facilitate that well thinking about basic needs and what have you and due to lack of time I'm not going to get really into that but what I will say um about cognitive strain is like you know if if you over if the agent is overloaded with with with stressors like prolonged stressors right and that contributes to to trauma and we could think of trauma as like being the result of this prolonged exposure to to stressors and and that's associated with the amplification and magnification of of cognitive strain um I would get more into um in into into uh into addiction and all that stuff and dealing with addiction but uh I think that's more more or less what I've I've uh thrown together so far in the body of my work I do however plan on talking about Free Will Dynamics so to summarize what I feel about uh what makes Free Will relevant um my point on Free Will is uh probably more on the compatible is point of view where um we consider degrees of freedom and willpower to uh align our perceptions actions and priors with regard to uh constraints in our environment that enables us to have to make choices reflective of how we feel with existing circumstances and how those circumstances align in the future and that moments of of of nondeterminism really are uh become resolved when uh choices reduce or evaluate to to how certain certain certain decisions play out and certain choices on line so uh non-deterministic uh non non moments of non-determinism become resolved into deterministic actions over time based on how choices play out reflective to uh present moments of circumstances which agents deal with that that carry over and align with future scenarios that the agent finds them themselves in so that that was another that that that's the last main that has to do with the the the section on Free Will that's the last main section and then there's the conclusion section of the paper of course I I also have a modeling section um I didn't expect to go over models today but um I kind of elaborated what my models would go over uh um but um I think that more or less um talks about that more or less covers uh you know there's a lot more I could talk about but more or less um covers U the work that I've been putting into this this specific overiew paper of habits and adaptive learning Dynamics with respect to uh agents agal system um yeah I I know I can go on and on and thanks for that I just noticed that uh message in the chat uh not terribly long ago but yeah well thank you Jesse let us take a deep breath okay yeah you covered a lot of territory I'll just read two questions from the live chat just to note them and then I'll just ask a final question and then you can give a few minute response and then we'll close it okay so Michael Lennon wrote how would he contrast the modification of individual habits versus habits between paired individuals versus habits of clusters of individuals and Michael's second question was how might he describe how a moment of Novel individual Insight becomes Collective Behavior change but just like to close it all out overall where are you going to take this work and H how do you intend to proceed it right um so reflecting on Michael's uh questions um so um let's address those another time I know that you could go into a lot of information on let's just go right to the next steps yeah yeah um so the next steps is um so this current paper is really an overview of um addressing what are what are habits what is what do habits have to do with adaptive learning and um how that ties with agal systems and in the conclusion section I'm going to bring about like what's going to happen with the the the next couple series of papers uh but really um I was originally going to go into the interventionary approach approaches geared towards adaptive learning but um so the next steps I want to do with this work is uh go into the interventionary approaches towards adaptive learning so okay how can we use technology how can we uh are there different modes in which we can better engage with information uh what about just how we engage the world around us like how like the food we eat like how do we like are there ways that we regulate food intake how about better understanding reasoning and emotion like how like is there some capacity in which we can better uh how we can how we can better reason with our emotions to more have more effective reasoning capacities to to deal with with circumstances largely dealing with non-triviality um so those are the those are the four next papers I consider the four Paradigm so the first Paradigm has to do with technological interventions like how can we utilize techn technology to better uh facilitate how we engage with the world around us and better benefit not only our own agency but each other's agency um and two would be U uh mode uh Paradigm two has to do with okay how can we better engage with certain objects in the world like like using tactile tools uh how can we use uh different objects to facilitate how we better engage with the world around us uh three dieting Dynamics how how can we leverage met abolism the most optimal fashion to to to to to leverage agency and for of course uh uh reasoning like is there some means in which we can better facilitate how we uh uh deal with emo affective processing and and how that can better be uh how reasoning can better uh leveraged with better understanding of deing with with leveraging effective processing um so that that's the next steps for me and those are going to be individual papers although I do have a uh I do plan on talking about the different brain regions so like a neuroscientific uh more of a neuro psychological uh approach in terms of and going talking about the different brain regions that are associated with uh habits and adaptive learning and agency but that would be a fifth paper um um okay last question how could people contact you or what would you look to ask people for or how they could get involved um uh I uh I do have like a business email but what I I'll I'll I'll I'll have like an email where that uh I'll provide a different email uh that I can have people reach out to me uh by it's it's a Google Mail account um so uh reach out to me um and hopefully um uh you know I I would love to try and collaborate with the right parties um and see where this work can go I I mean I have a lot of uh confidence in terms of especially with uh the system model that I'm designing in terms of technological interventions I would love to talk to people about that uh because that's the next main step I'm going with is okay well there's a problem with how we interface techn like with social media and like just a lot of platform like just a lot of basic Technologies right now that really compromises how we engage with you in with the world like it's addictive like like social media is addictive like people play mobile games on their phone right right and that people can waste hours doing all that stuff so how can we better develop systems that complement rather than compromise our behavior um I think that's really important to figure out um and I would love to talk to more people about that um awesome well you covered a ton of scope I hope that people who are studying active inference see a lot of the connections in the terms and topics and systems and good luck in your learning and research thank you I appreciate it peace Jesse bye for